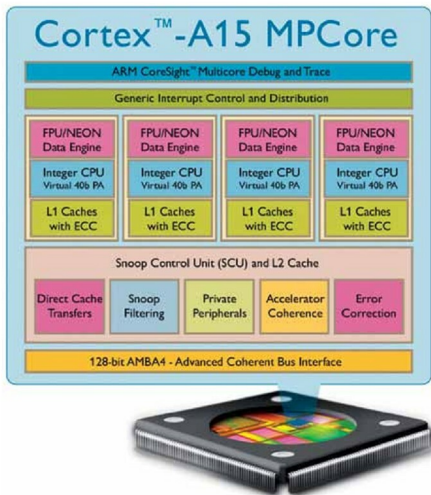




and much faster set of boot code like coreboot.

Steve Goodrich, principal, Sage Electronic Engineering, explains, "You wouldn't use coreboot in every case where you need a BIOS. A BIOS is typically more flexible and intended to work with systems on which the hardware can vary from one boot to another, while coreboot is targeted at systems on which the hardware does not vary, such as an embedded device."

Commercial-off-the-shelf (COTS) board. Software development on x86 platforms has been quite convenient. However, "The complexity of hardware design and competition from Taiwan prohibits designing of the hardware locally," shares Sinha. He explains that since x86 platforms have been very popular in digital signage, industrial and other such applications,

Taiwanese are offering these products at a very low cost. For example, you can buy a complete motherboard for less than Rs 4000. However, if you have to build it locally, you won't be able to match the price-point. They have a very good infrastructure to lower the costs.

Adding to the advantages of COTS x86-boards, Sharma shares, "These are extremely useful when you have to deliver a system in a short period and the hardware cycles are unacceptable, volumes are low and the non-recurring cost of development cannot be offset." He also feels that time-to-market pressures obviate the need for x86 in such a situation.

The disadvantage of COTS boards is their short life of 3-6 months causing you to retest your solutions every time you try a new board. Rosenboim

shares, "Most often, an x86 board is chosen for its low cost and low volume, but it also has a short product life, forcing you to replace discontinued boards with new ones that cause unplanned and non-budgeted re-engineering costs and associated risks."

Applications for x-86

For PC-like performance and high computing capabilities without bothering much about the power efficiency, system developers would prefer to have the x86-based solutions. Industrial motherboards used in automation and other applications that are expected to run a heavy operating system like Windows use x86-based platform.

Sharma shares, "When the list of peripherals that the user is looking for in the system is large, and he wants to avoid the cost of developing all those device drivers, he prefers to use an x86-based solution in order to leverage the already existing set of device drivers and peripherals." However, a generic x86 is usually lesser-equipped and more expensive than other options available in the market.

The collection of peripherals gives a personality to the chip. "Today, processors are coming as system-on-chip with a large number of peripherals built into the die, giving a much cheaper, more appropriate alternative to the final product," says Sharma.

Test and measurement equipment. Test and measurement equipment are highly complex embedded systems. These demand precision and accuracy, which is mostly real-time. There are a number of oscilloscopes using x86, as they have an inbuilt Windows XP operating system running beneath.

"x86 architecture-based Atom is also used in data-acquisition products. This is because readymade original equipment manufacturer (OEM) boards are easily available which meet the performance requirements," Sinha shares.

Networking and communication. Networking and communication is a matured and standardised application area. There are standards and protocols which